

Supplemental Material for “Interband Pairing Signatures in the Superfluid Response of Magic-Angle Twisted Bilayer Graphene”

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TABLE I. Default parameters used for the mechanism scans.

quantity	value
twist angle	$\theta = 1.05^\circ$
Dirac velocity	$v_F = 2.135 \text{ eV \AA}$
AA tunneling	$w_0 = 87.2 \text{ meV}$
AB tunneling	$w_1 = 109.0 \text{ meV}$
reciprocal cutoff	$N_{\text{shell}} = 3$
retained bands	$n_{\text{keep}} = 6$
momentum grid	$nk = 7$ for dense maps, $nk = 9, 11, 13$ for key-point checks
gap scale	$\Delta_0 = 1 \text{ meV}$
velocity finite difference	$dk = 10^{-6} \text{ \AA}^{-1}$

I. SCOPE OF THE SUPPLEMENTAL MATERIAL

This Supplemental Material documents the executable numerical workflow behind the PRB manuscript. It is organized as a reproducibility and self-audit record: the Bistritzer-MacDonald (BM) model conventions [1], the orbital-projected pairing construction, the finite-band response formula, the figure and table provenance, and the historical PRB benchmark reconstruction are recorded separately.

The main text makes a conservative mechanism claim: an orbital-projected interband-pairing direction produces a reproducible, normalization-conditioned change in the normalized finite-band superfluid response. The present Supplemental Material does not promote the raw response units to final experimental stiffness values. This choice is motivated by the distinction between robust quantum-geometric mechanism diagnostics [2–5] and direct quantitative comparison with recent stiffness experiments [6, 7].

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II. BM MODEL AND NUMERICAL PARAMETERS

The normal-state calculation uses the single-valley BM continuum Hamiltonian with orbital basis

$$(tA, tB, bA, bB), \quad (1)$$

where t, b label layers and A, B label sublattices. Unless explicitly stated otherwise, the production mechanism scans use the parameters in Table I.

The moire unit-cell area and reciprocal scale used in the unit audit are

$$A_M = 1.5606 \times 10^4 \text{ \AA}^2, \quad G_M = 0.054047 \text{ \AA}^{-1}. \quad (2)$$

The raw response conversion used in the audit is

$$D[\text{eV \AA}^2] = \frac{D_{\text{raw}}[\text{meV \AA}^2]}{1000}, \quad (3)$$

with all remaining spin, valley, or convention factors kept explicit rather than absorbed into a hidden prefactor.

A. Reporting Convention for Units and Filling

Table II states the reporting convention used in the present manuscript. The main figures use normalized response ratios, while absolute $\text{eV \AA}^2$ values are restricted to the self-audit and historical benchmark reconstruction. This is the final observable policy for the current mechanism-paper scope.

TABLE II. Reporting convention for response and filling quantities.

quantity	present use	excluded interpretation
raw response	internal finite-band diagnostic	physical stiffness
$D_{\text{iso}}(\eta)/D_{\text{iso}}(0) - 1$	claim-bearing mechanism observable	absolute stiffness
$\text{eV \AA}^2$ per flavor	benchmark audit only	experimental comparison
ν_{proxy}	retained-band occupancy label	calibrated carrier density

The retained-band filling proxy is computed as

$$\nu_{\text{proxy}} = g \left[\left\langle N_{\text{keep}}^{-1} \sum_n \Theta(\mu - \varepsilon_{n\mathbf{k}}) \right\rangle_{\mathbf{k}} - \frac{1}{2} \right], \quad (4)$$

with $g = 4$ used as a spin-valley counting factor in the proxy. For the $n_{\text{keep}} = 6$ scans this is not identical to a calibrated experimental filling factor; it is used only to order the chemical-potential scans. The central flat-band crosswalk instead counts the two central BM bands as

$$\nu_{\text{flat}} = g \left[\left\langle \sum_{n \in \text{central pair}} \Theta(\mu - \varepsilon_{n\mathbf{k}}) \right\rangle_{\mathbf{k}} - 1 \right], \quad (5)$$

which ranges from -4 to $+4$ for an empty-to-full central two-band subspace when $g = 4$. On the $nk = 7$, $N_{\text{shell}} = 3$ mesh used for the dense response scan, the central-pair energy window is $[-1.051, 4.134]$ meV. The resulting crosswalk is shown in Table III. It should be read as a flat-band counting reference for the present BM calculation, not as a device-specific carrier-density calibration.

TABLE III. Crosswalk between the retained-band filling proxy and central two-flat-band filling for the dense $nk = 7$, $n_{\text{keep}} = 6$ scan.

μ (meV)	ν_{proxy}	ν_{flat}	occupied flat bands/flavor
-5	-1.483	-4.000	0.000
-4	-1.333	-4.000	0.000
-3	-1.061	-4.000	0.000
-2	-0.857	-4.000	0.000
-1	-0.653	-3.837	0.041
0	-0.381	-2.286	0.429
1	-0.054	-0.327	0.918
2	0.122	0.735	1.184
3	0.395	2.367	1.592
4	0.667	3.918	1.980
5	0.857	4.000	2.000

III. ORBITAL-PROJECTED PAIRING CONSTRUCTION

The pairing direction is specified first in the layer-sublattice orbital basis,

$$\Delta_{\text{orb}}(\eta) = \Delta_0 \mathcal{N}_\eta(M_0 + \eta M_1), \quad (6)$$

TABLE IV. Raw valley-minus sewing diagnostic. The independently diagonalized valley-minus subspace at $-\mathbf{k}$ is compared with the target $U_+^*(\mathbf{k})$ on an $nk = 3$ grid. Small singular values and large alignment errors indicate that the raw valley-minus eigenvectors should not be used directly to define W_{inter} .

n_{keep}	alignment error	mean minimum singular value	minimum singular value
2	1.3895	0.0267	0.0218
4	1.3782	0.0201	0.0043
6	1.3760	0.0095	0.0016

with

$$M_0 = \tau_0 \sigma_0, \quad M_1 = \tau_x \sigma_x. \quad (7)$$

The two normalization controls used in the manuscript are

$$\text{fixed Frobenius:} \quad \|M_0 + \eta M_1\|_F = \|M_0\|_F, \quad (8)$$

$$\text{fixed } \Delta_0 : \quad \mathcal{N}_\eta = 1. \quad (9)$$

The band-basis pairing matrix is

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k}) \Delta_{\text{orb}} U_-^*(-\mathbf{k}). \quad (10)$$

The working valley convention in the present diagnostic is the time-reversal sewn proxy

$$U_-(-\mathbf{k}) = U_+^*(\mathbf{k}). \quad (11)$$

The production projection is therefore

$$\Delta_{\text{band}}(\mathbf{k}) = U_+^\dagger(\mathbf{k}) \Delta_{\text{orb}} U_+(\mathbf{k}). \quad (12)$$

This convention is useful for isolating an orbital-defined interband-pairing signature. It also avoids a known gauge artifact: independently diagonalized valley-minus eigenvectors are not automatically in the same time-reversal sewing convention as the valley-plus eigenvectors. Table IV summarizes the raw-basis diagnostic.

This is still a diagnostic convention rather than a full independent two-valley implementation. Valley-asymmetric perturbations, strain, or valley-dependent interactions would require revisiting this sewing choice.

The projected interband-pairing weight is

$$W_{\text{inter}} = \frac{\sum_{n \neq m} |\Delta_{nm}|^2}{\sum_{n,m} |\Delta_{nm}|^2}. \quad (13)$$

For the dense $nk = 7$, $n_{\text{keep}} = 6$ scan, $W_{\text{inter}} \simeq 0.108$ at $\eta = 1$.

IV. FINITE-BAND RESPONSE AND CHANNEL SPLIT

For a retained band subspace, the finite-band BdG Hamiltonian is

$$H_{\text{BdG}}(\mathbf{k}) = \begin{pmatrix} \varepsilon(\mathbf{k}) - \mu & \Delta_{\text{band}}(\mathbf{k}) \\ \Delta_{\text{band}}^\dagger(\mathbf{k}) & -[\varepsilon(\mathbf{k}) - \mu] \end{pmatrix}. \quad (14)$$

The zero-temperature static current-current response is evaluated as

$$\Lambda_{\alpha\beta} = \sum_{a \neq b} \frac{\langle a | J_\alpha | b \rangle \langle b | J_\beta | a \rangle (f_a - f_b)}{E_b - E_a}. \quad (15)$$

The band-basis velocity is split into diagonal and off-diagonal parts,

$$v_\alpha^{\text{band}} = v_\alpha^{\text{intra}} + v_\alpha^{\text{inter}}. \quad (16)$$

This produces conventional, geometric, and cross-channel response diagnostics. The primary normalized response map uses

$$\delta D_{\text{iso}}(\mu, \eta) = \frac{D_{\text{iso}}(\mu, \eta)}{D_{\text{iso}}(\mu, 0)} - 1, \quad D_{\text{iso}} = \frac{D_{xx} + D_{yy}}{2}. \quad (17)$$

V. FIGURE AND TABLE PROVENANCE

Table [V](#) summarizes the current figure and table provenance. The full machine-readable record is maintained in `notes/figure_table_provenance_2026-06-09.md`.

The dense response data are reproduced by

```
python3 scripts/run_mu_response_scan.py \
--nk 7 --n-shell 3 --n-keep 6 \
--mus -5 -4 -3 -2 -1 0 1 2 3 4 5 \
--etas 0 0.25 0.5 0.75 1 \
--output data/processed/mu_eta_response_scan_nk7_nkeep6.csv
```

TABLE V. Manuscript artifact provenance. CSV values used in the manuscript tables are checked by `scripts/verify_manuscript_tables.py`.

artifact	data source	script	check
Frobenius heatmap	dense scan CSV	heatmap plotter	figure file exists
fixed- Δ_0 heatmap	dense scan CSV	heatmap plotter	figure file exists
dense response table	eta-summary CSV	eta summarizer	table verifier
$nk = 9$ validation table	key-point summary CSV	eta summarizer	table verifier
PRB audit table	reconstruction CSVs	audit scripts	table verifier

and summarized by

```
python3 scripts/summarize_eta_effect.py \
    data/processed/mu_eta_response_scan_nk7_nkeep6.csv \
    --output data/processed/mu_eta_response_scan_nk7_nkeep6_eta1_summary.csv
```

The table verifier is

```
python3 scripts/verify_manuscript_tables.py
```

VI. KEY-POINT MOMENTUM-GRID CONVERGENCE

The dense maps in the main text use $nk = 7$, while selected key points were checked at $nk = 9$, $nk = 11$, and $nk = 13$. The higher-grid scans use

```
python3 scripts/run_mu_response_scan.py \
    --nk 11 --n-shell 3 --n-keep 6 \
    --mus -4 0 2 4 --etas 0 0.5 1 \
    --output data/processed/mu_response_scan_nk11_nkeep6_keypoints.csv
```

```
python3 scripts/run_mu_response_scan.py \
    --nk 13 --n-shell 3 --n-keep 6 \
    --mus -4 0 2 4 --etas 0 0.5 1 \
    --output data/processed/mu_response_scan_nk13_nkeep6_keypoints.csv
```

and is summarized with `scripts/summarize_eta_effect.py`. The convergence figure is generated with

```
python3 scripts/plot_nk_convergence_keypoints.py
```

Figure 1 and Table VI compare the normalized $\eta = 1$ response relative to $\eta = 0$. The fixed-Frobenius response remains positive at all four key chemical potentials. The fixed- Δ_0 response remains small and sign-sensitive, consistent with the main-text interpretation that the interband-pairing signal is normalization-conditioned.

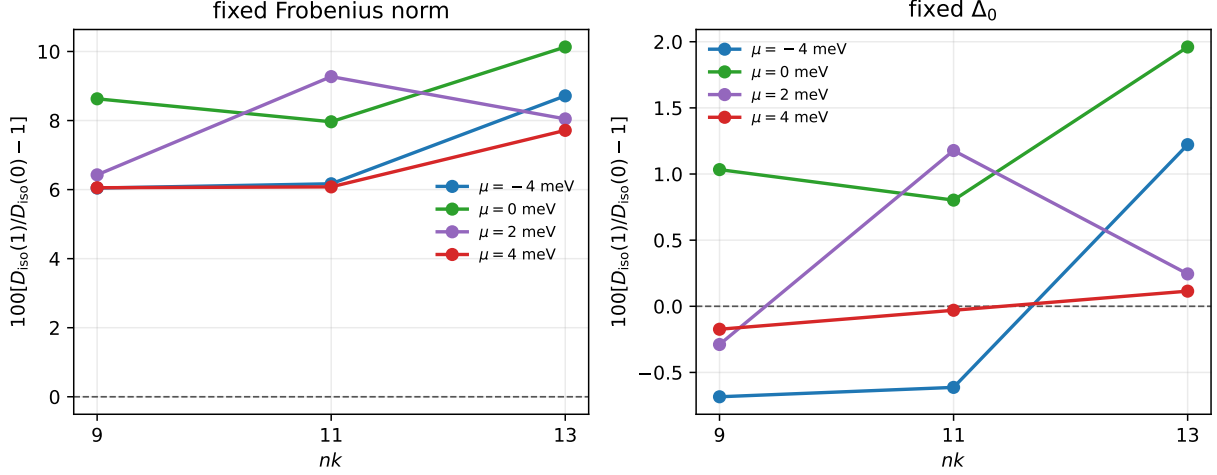


FIG. 1. Selected-grid convergence of the normalized $\eta = 1$ response. The left panel shows the fixed-Frobenius comparison used for the main mechanism claim; the right panel shows the fixed- Δ_0 control. The plotted values are drawn from the same summary data tabulated in Table VI.

TABLE VI. Key-point momentum-grid check for $D_{\text{iso}}(\eta = 1)/D_{\text{iso}}(\eta = 0) - 1$.

mode, μ (meV)	$nk = 9$	$nk = 11$	$nk = 13$	$13 - 9$
fixed Δ_0 , -4	-0.0068	-0.0061	+0.0122	+0.0191
fixed Δ_0 , 0	+0.0103	+0.0080	+0.0196	+0.0093
fixed Δ_0 , 2	-0.0029	+0.0118	+0.0024	+0.0053
fixed Δ_0 , 4	-0.0017	-0.0003	+0.0011	+0.0029
fixed Frobenius, -4	+0.0604	+0.0617	+0.0872	+0.0267
fixed Frobenius, 0	+0.0863	+0.0796	+0.1013	+0.0150
fixed Frobenius, 2	+0.0643	+0.0927	+0.0805	+0.0162
fixed Frobenius, 4	+0.0605	+0.0608	+0.0772	+0.0166

Across the three key-point grids the fixed-Frobenius response remains positive, with the largest $nk = 13 - nk = 9$ shift about 2.67 percentage points. Therefore the current manuscript claims sign and order-of-magnitude stability of the normalized mechanism signal, not a strict continuum-limit extrapolation of the tabulated values.

As an additional finite-grid audit, the same values are fitted to

$$y(nk) = y_0 + a/nk^2, \quad y = 100 [D_{\text{iso}}(\eta = 1)/D_{\text{iso}}(\eta = 0) - 1]. \quad (18)$$

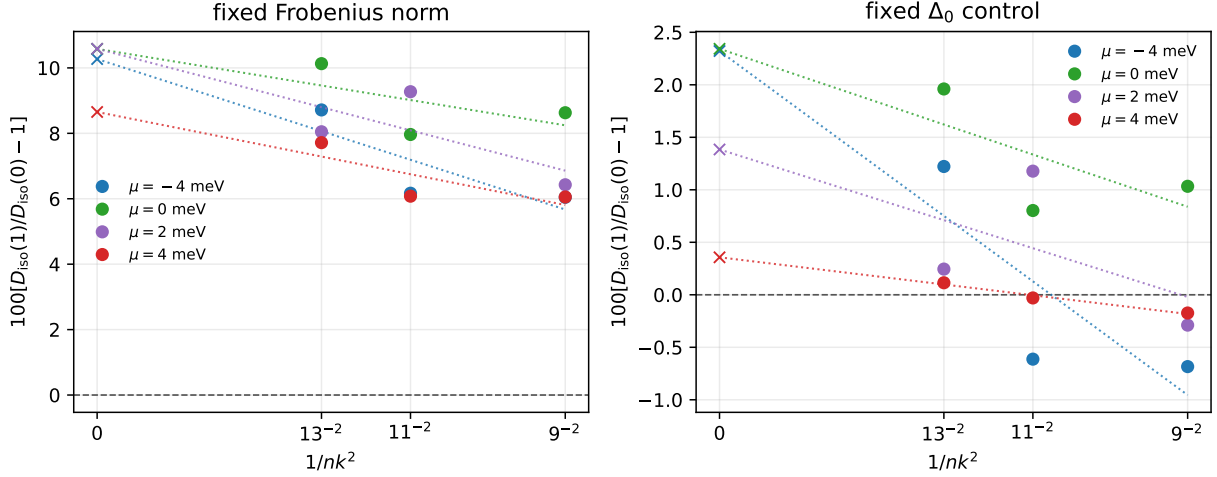


FIG. 2. Finite-grid trend audit for the selected $nk = 9, 11, 13$ key points. Markers at nonzero $1/nk^2$ are computed values, dotted lines are linear $1/nk^2$ fits, and crosses at zero are fitted intercepts. The fixed-Frobenius trend remains positive for all four chemical potentials; the fixed- Δ_0 control remains much smaller and sign-sensitive on the measured grids.

TABLE VII. Finite-grid trend audit for the selected key points. Values are in percent. The $1/nk^2$ intercept is the plotted trend marker, while the $1/nk$ intercept is a sensitivity check.

mode, μ (meV)	min	max	$1/nk^2$ int.	$1/nk$ int.	sign
fixed Δ_0 , -4	-0.68	1.22	2.32	4.81	mixed
fixed Δ_0 , 0	0.80	1.96	2.34	3.53	positive
fixed Δ_0 , 2	-0.29	1.18	1.38	2.20	mixed
fixed Δ_0 , 4	-0.17	0.11	0.36	0.74	mixed
fixed Frobenius, -4	6.04	8.72	10.27	13.77	positive
fixed Frobenius, 0	7.96	10.13	10.58	12.48	positive
fixed Frobenius, 2	6.43	9.27	10.58	12.94	positive
fixed Frobenius, 4	6.05	7.72	8.65	10.82	positive

Figure 2 shows the measured points, the $1/nk^2$ trend lines, and the intercept markers at $1/nk^2 = 0$. Table VII also lists a $1/nk$ intercept as a simple sensitivity check. With only three grids, these intercepts are an audit of trend stability rather than a final continuum extrapolation. The fixed-Frobenius intercepts remain positive in both trend coordinates,

TABLE VIII. Targeted $nk = 15$ spot check for representative key points.

mode, μ (meV)	$nk = 13$	$nk = 15$	$15 - 13$	$W_{\text{inter}}(15)$
fixed Δ_0 , 0	+0.0196	+0.0085	-0.0111	0.1076
fixed Δ_0 , 2	+0.0024	+0.0071	+0.0046	0.1076
fixed Frobenius, 0	+0.1013	+0.0897	-0.0116	0.1076
fixed Frobenius, 2	+0.0805	+0.0865	+0.0060	0.1076

whereas the fixed- Δ_0 measured-grid signs remain mixed except at $\mu = 0$.

A targeted higher-grid spot check was also run at $nk = 15$ for $\mu = 0$ and 2 meV, using only $\eta = 0$ and $\eta = 1$. This is not a full dense-grid replacement, but it tests whether the two central representative key points remain in the same response range on a denser mesh. Table VIII shows that the fixed-Frobenius response remains positive and close to the $nk = 13$ values, while the fixed- Δ_0 control remains below one percent at both chemical potentials.

VII. HISTORICAL PRB BENCHMARK RECONSTRUCTION

The production interband-pairing scan and the historical PRB table reconstruction are kept separate. The best simple unified reconstruction route for the old uniform- s benchmark is

$$D_{\text{conv}} = 2D_{\text{intra},\tau_z}, \quad D_{\text{geom}} = D_{\text{inter},\tau_z}. \quad (19)$$

For $n_{\text{keep}} = 6$, this gives $D_{\text{total}} = 126.66 \text{ eV } \text{\AA}^2$ versus the historical target $129.30 \text{ eV } \text{\AA}^2$, and $D_{\text{geom}} = 75.22 \text{ eV } \text{\AA}^2$ versus $75.30 \text{ eV } \text{\AA}^2$.

The old $n_{\text{keep}} = 2$ endpoint is more convention-sensitive. The closest audited reconstruction combines a Gamma-centered mesh, central-pair band selection, an all- τ_z current convention, a factor of two in the intraband term, and a full curvature-like conventional correction. This gives

$$D_{\text{total}} = 66.18 \text{ eV } \text{\AA}^2 \quad \text{versus} \quad 67.50 \text{ eV } \text{\AA}^2. \quad (20)$$

Because this full-curvature correction does not work uniformly at $n_{\text{keep}} = 6$, the reconstruction is treated as a historical audit rather than a production convention.

VIII. CLAIM SCOPE AND EXCLUDED INTERPRETATIONS

The manuscript is framed as a normalized-response mechanism paper. The text-level claim-scope audit is generated with

```
python3 scripts/audit_claim_scope.py
```

and the observable-policy audit is generated with

```
python3 scripts/audit_observable_policy.py
```

The audit outputs are

```
data/processed/claim_scope_audit.csv
```

```
data/processed/observable_policy_audit.csv
```

They are not numerical validation tools; they check that the manuscript retains the boundary statements needed to avoid overinterpreting the finite-band diagnostic calculation. Table IX summarizes the current claim scope.

TABLE IX. Allowed manuscript claims and excluded interpretations.

topic	allowed in the present manuscript	excluded interpretation
normalized response mechanism signature		absolute stiffness enhancement
fixed- Δ_0 control	weak/sign-sensitive control	universal enhancement
momentum grids	selected sign/scale stability	continuum-limit value
filling variable	central flat-band crosswalk	device density calibration
absolute units	benchmark self-audit only	experimental comparison
valley partner	tr_sewn diagnostic convention	valley-asymmetric calculation

IX. CURRENT LIMITATIONS AND NEXT VALIDATION GATES

The current submission-critical gaps are:

1. a full continuum-grid extrapolation beyond the present selected $nk = 9$, $nk = 11$, $nk = 13$ trend audit and targeted $nk = 15$ spot check;

2. a fully independent two-valley implementation before studying valley-asymmetric perturbations;
3. a final absolute stiffness convention only before any direct experimental comparison;
4. a device-level filling calibration beyond the central-flat-band crosswalk reported here.

These limitations do not invalidate the normalized-response mechanism claim, but they define the boundary of the present PRB mechanism submission. Direct experimental-comparison claims require further convergence, calibration, and absolute-convention work.

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DECLARATIONS

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Conflict of interest: The author declares no conflict of interest.

Data availability: All data and code supporting this work are available from the corresponding author upon reasonable request.

Ethics approval: Not applicable.

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